KisanAI

Har Khet Mein AI — AI in Every Farm

Problem Statement 5 — Domain-Specialized AI Agents with Compliance Guardrails

ET AI Hackathon 2026 | Built by Mahendra

🔗 **kisansathi-smart-farm.lovable.app**

1. System Overview

KisanAI is a multi-agent AI powered farming super app designed for India's 140 million smallholder farmers. It integrates AI crop disease detection, live mandi market prices, weather intelligence, land and machinery marketplaces, farm work job matching, soil health analysis, and organic farming support — all in one unified platform available in 6 Indian languages.

The system is built on a React TypeScript frontend with Supabase Edge Functions as the backend. Anthropic Claude (claude-sonnet-4-20250514) powers all AI features including natural language farming advice, vision-based disease diagnosis, soil analysis, and personalized government scheme recommendations.

2. Problem We Are Solving

Problem	Scale	KisanAI Solution
No expert farming advice	140M farmers	Claude AI farming assistant in 6 languages
Cannot identify crop diseases	30% crop loss	Photo upload → instant diagnosis + medicine
No fair market price info	Millions exploited	Live mandi prices per KG from govt API
Cannot find farm machinery	Seasonal shortage	Location-based machinery rental marketplace
Land not being farmed	Millions of acres idle	Land owner to farmer direct connect
Cannot find farm workers	Harvest time crisis	Nearby job notifications within 30km

Problem	Scale	KisanAI Solution
No soil health guidance	Declining soil fertility	AI analysis + Agriculture Officer alert
No organic farming support	Chemical dependency	Organic marketplace + education

3. System Architecture

👨 🌾 **FARMER INPUT — Text / Photo / Voice in Any Indian Language**

▼

🤖 **ORCHESTRATOR AGENT — Claude AI — Intent Classification and Agent Routing**

▼

🔬 Disease Agent	📈 Market Agent	🌤️ Weather Agent	🧪 Soil Agent
Photo → Diagnosis Medicine + Shop	Live Mandi Prices AI Market Advice	GPS Weather Farming Tips	Soil Score Crop Advice

▼

🛡️ **COMPLIANCE GUARDRAIL — Banned Pesticide Check — CIB&RC India Guidelines — Audit Log**

▼

📱 **FARMER GETS SAFE RESPONSE — In Their Own Language — With Action Buttons**

4. Complete Feature List

#	Feature	AI Used	APIs Used	Description
1	AI Farming Chat	Claude AI	Anthropic API	Answers any farming question in 6 Indian languages. Personalized using farmer profile. Suggests govt schemes.
2	Disease Detector	Claude Vision	Anthropic API	Photo of sick plant → disease name, severity, medicine with dosage, treatment steps, nearest

#	Feature	AI Used	APIs Used	Description
				pesticide shop.
3	Live Market Prices	Claude AI	data.gov.in API	Real mandi prices per KG for 60+ crops. Vegetables, fruits, grains, flowers, pulses, spices. State-wise filtering.
4	Weather Widget	Claude AI	OpenWeatherMap	GPS-based weather. Temperature, humidity, rain chance. AI farming tip based on weather conditions.
5	Land Connect	None	Supabase + GPS	Landowners post available land. Farmers find land sorted by distance. Direct call and WhatsApp contact.
6	Machinery Rental	None	Supabase + GPS	Machine owners post availability. Farmers find machines by distance and date. Solves seasonal shortage.
7	Farm Work Jobs	None	Supabase + GPS	Farmers post work. Workers see jobs within 30km. One-tap WhatsApp apply. Notification badge for new jobs.
8	Soil Health	Claude AI	Anthropic API	AI soil analysis with score out of 10. Crop recommendations. Agriculture Officer alert with reference number.
9	Organic Market	Claude AI	Supabase + Anthropic	Buy and sell gobar, compost, organic fertilizers. Learn Organic tab with complete guide from Claude AI.
10	Farmer Profile	Claude AI	localStorage	Complete farmer profile used to personalize all AI responses. Farm size, crops, location, farming type.
11	Govt Schemes	Claude AI	Anthropic API	Top 5 personalized government scheme recommendations based on farmer profile. PM Kisan, KCC, PMFBY.
12	Notifications	None	Supabase	New nearby jobs, machinery, land listings, price alerts, soil report status updates.

5. AI Agent Roles and Responsibilities

Agent	Trigger	Input	Output	Compliance Check

Agent	Trigger	Input	Output	Compliance Check
Orchestrator	Every farmer query	Farmer message + profile + language	Routes to correct specialist agents	None — routing only
Disease Agent	Photo uploaded	Crop photo as base64	Disease, medicine, dosage, prevention	Banned pesticide check — HARD BLOCK
Weather Agent	Home screen load	GPS coordinates	Weather data + farming tip	None required
Market Agent	Market tab open	State + category filter	Live prices + AI market advice	None required
Soil Agent	Soil description or photo	Text or image input	Soil score + crop recommendations	Chemical recommendation check
Scheme Agent	Profile complete	Farmer profile data	Top 5 personalized schemes	None required
Compliance Layer	Every AI output	Raw Claude response	Safe filtered response	Blocks 8 banned pesticides

6. Agent Communication Flow

Step	Actor	Action	Detail
1	Farmer	Sends query or photo	Text, image, or voice via app interface
2	Orchestrator	Classifies intent	Claude AI determines which agents are needed
3	Specialist Agents	Execute in parallel	Disease, Weather, Market, Soil agents run simultaneously
4	Compliance Layer	Validates all outputs	Checks every response against banned substance list
5	Audit Logger	Records everything	Timestamp, agent, input, output, compliance status saved to Supabase
6	Orchestrator	Synthesizes response	Combines agent outputs into single farmer-friendly answer

Step	Actor	Action	Detail
7	Farmer	Receives safe response	In their selected language with action buttons

7. Compliance and Safety Guardrails

Every disease and soil medicine recommendation passes through the Compliance Guardrail Layer before reaching the farmer. This is a HARD BLOCK system — if any banned substance is detected, it is automatically replaced with a safe approved alternative.

Banned Substance	Reason Banned	Safe Alternative Provided
Endosulfan	Highly toxic — banned India 2011	Neem-based insecticide Azadirachtin
Monocrotophos	Extremely hazardous — WHO Class Ib	Spinosad or Emamectin benzoate
Methyl Parathion	Acutely toxic — banned India	Chlorantraniliprole or Flubendiamide
Carbofuran	High toxicity to birds and humans	Thiamethoxam or Imidacloprid
Phorate	Highly hazardous organophosphate	Fipronil or Chlorpyrifos alternatives
Triazophos	Banned for vegetables India 2013	Profenofos or Lambda-cyhalothrin
Dimethoate	Restricted use India	Acetamiprid or Thiamethoxam
Lindane	Persistent organic pollutant	Deltamethrin or Cypermethrin

Compliance badge shown on every AI recommendation: 'Recommendations follow CIB and RC India approved guidelines'

Every AI recommendation logged to Supabase ai_logs table with timestamp, agent name, input summary, output summary, and compliance status.

8. Database Schema

Table	Key Fields	Purpose
land_listings	id, owner_name, village, district, lat, lng, acres, soil_type, rent, phone	Land owner to farmer matching
machinery_listings	id, machine_type, brand, lat, lng, available_from, available_to, price_per_day, phone	Machinery rental marketplace
farm_jobs	id, work_type, crop, start_date, lat, lng, workers_needed, wage_per_day, meals, phone	Farm work job matching
organic_listings	id, product_type, quantity, unit, price, lat, lng, phone, photo_url	Organic compost marketplace
authority_alerts	id, farmer_name, village, phone, problem_description, lat, lng, status, reference_number	Soil issue reporting to agriculture dept
ai_logs	id, agent_name, input_summary, output_summary, compliance_status, timestamp	Immutable AI decision audit trail
notifications	id, message, type, read, created_at	In-app notification storage
price_alerts	id, crop_name, target_price, phone, created_at	Farmer price alert subscriptions

9. Technology Stack

Layer	Technology	Purpose
Frontend	React + TypeScript	Farmer-facing web application
Backend	Supabase Edge Functions (Deno)	Serverless API functions for all AI calls
Database	Supabase PostgreSQL	All listings, logs, notifications, alerts
Storage	Supabase Storage	Crop photos, machine photos, profile photos
AI Brain	Anthropic Claude claude-sonnet-4-20250514	Chat, disease detection, soil analysis, market advice
Weather Data	OpenWeatherMap API	Real-time GPS-based weather for farmers
Market Data	data.gov.in Agmarknet API	Live daily mandi prices from all Indian APMCs

Layer	Technology	Purpose
Maps	Google Maps URL Integration	Nearest pesticide shop finder
Hosting	Lovable Platform	Automatic deployment and hosting
Languages	i18n Translation Object	English Hindi Kannada Telugu Tamil Malayalam

10. API Integrations

API	Endpoint	Used For	Cost
Anthropic Claude	api.anthropic.com/v1/messages	All AI features — chat, disease, soil, schemes	Paid — approx Rs 50 for entire hackathon
OpenWeatherMap	api.openweathermap.org/data/2.5/weather	GPS weather + farming tips	Free tier — 1000 calls per day
data.gov.in	api.data.gov.in/resource/agmarknet	Live mandi prices all crops all states	Free — open government data
Google Maps	google.com/maps/search URL	Nearest pesticide shop finder	Free — URL redirect only
Supabase	project.supabase.co	Database, storage, edge functions	Free tier sufficient
Kisan Helpline	1800-180-1551 tel link	Direct call to government helpline	Free — government service

11. Multi-Language Support

KisanAI supports 6 Indian languages. Language selection is saved to localStorage and passed to every Claude API call. All UI elements, buttons, labels, error messages, and AI responses are delivered in the farmer's chosen language.

Language	Code	UI Translation	AI Response
English	en	✅ Complete	✅ Complete
हिंदी Hindi	hi	✅ Complete	✅ Complete
ಕನ್ನಡ Kannada	kn	✅ Complete	✅ Complete
తెలుగు Telugu	te	✅ Complete	✅ Complete
தமிழ் Tamil	ta	✅ Complete	✅ Complete
മലയാളം Malayalam	ml	✅ Complete	✅ Complete

12. Quantified Impact Model

Assumptions: Average Indian small farmer, 1.1 hectare farm, annual income Rs 1,20,000. Based on published agricultural research and ICAR studies.

Impact Area	Mechanism	Saving Per Farmer Per Year
Better crop selection	Soil-matched crop choice improves yield by 10 percent	Rs 12,000
Market price optimization	Sell at right APMC at right time — 15 percent higher price	Rs 18,000
Early pest detection	Catch disease early — reduce crop loss by 8 percent	Rs 9,600
Weather-guided irrigation	Avoid over or under watering — save 20 percent input cost	Rs 6,000
Total per farmer per year		Rs 45,600
At 1 lakh farmers	Conservative adoption scenario	Rs 456 Crore aggregate income gain
India potential	140 million farmers total addressable	Rs 63,840 Crore potential impact

13. Error Handling and Self-Correction

Failure Scenario	Detection	Recovery Action
Claude API 429 rate limit	HTTP 429 response	Auto wait 3 seconds and retry once. Show typing animation during wait.
Claude API no credits	HTTP 401 response	Show friendly message: Please try again in a moment
Weather API timeout	Timeout after 5 seconds	Show last cached weather data with staleness warning
Market API empty data	Empty records array	Show realistic static fallback prices for 60 crops with note
GPS location denied	GeolocationPositionError	Show manual location search box for farmer to type location
Banned pesticide detected	String match against banned list	HARD BLOCK — replace with safe alternative automatically
Image too large for API	File size check before upload	Auto compress image to under 5MB before sending to Claude
Supabase connection fail	Network error catch	Show offline mode with cached listings data

14. Auditability and Compliance Trail

Every AI decision made by any agent is recorded in the Supabase ai_logs table with the following fields:

- Timestamp UTC — exact time of recommendation
- Agent Name — which agent made the decision
- Input Summary — sanitized version of farmer input
- Output Summary — what was recommended
- Compliance Status — PASS, WARN, or BLOCK
- Blocked Substances — list of any substances that were blocked
- Safe Alternatives — what was substituted if blocked
- Language — which language response was delivered in

Agriculture Officer Alert System: When farmer submits soil report, a record is created in authority_alerts table with status PENDING and a unique 8-digit reference number. The farmer

receives confirmation that an officer will contact them within 3 working days.

15. Problem Statement 5 Alignment

PS5 Requirement	KisanAI Implementation	Status
Domain expertise depth	Claude AI trained with deep Indian agriculture knowledge across all crops, soil types, pests, and schemes	✅ Fully met
Compliance and guardrail enforcement	Hard block on 8 banned Indian pesticides with automatic safe alternative substitution	✅ Fully met
Edge case handling	Offline mode, API fallbacks, GPS denied fallback, low confidence escalation to KVK	✅ Fully met
Full task completion	Farmer uploads photo → gets disease → gets medicine → finds nearest shop → gets directions. Complete journey.	✅ Fully met
Auditability of every agent decision	Complete immutable audit trail in Supabase ai_logs with compliance status for every recommendation	✅ Fully met
Multi-agent coordination	Orchestrator routes to Disease, Weather, Market, Soil, Scheme agents running in parallel	✅ Fully met